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Centro Occidental Polangui, Albay



IT112 WEB SYSTEM AND TECHNOLOGY

Phase 7-Final Project

THE SURVIVORS

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Title

Smart Attendance System with Auto-Time Classification

I. Introduction

This final project presents the development of the Smart Attendance System with Auto-Time Classification, a web-based platform designed to improve the way schools manage student attendance. Traditional attendance methods are commonly done manually using paper records or spreadsheets, which can lead to inaccurate data, time-consuming processes, and difficulty in monitoring student attendance efficiently. To solve these challenges, the system provides an automated and organized solution that records attendance digitally and accurately.

The system allows students to check in using their student ID while automatically recording the exact date and time of attendance. Based on predefined time rules, the system instantly classifies attendance as Early, On-Time, Late, or Absent. Teachers can easily monitor attendance records, while administrators can review reports and attendance trends. By using a user-friendly web-based platform, the Smart Attendance System helps schools improve efficiency, reduce manual errors, and maintain secure attendance records that are accessible anytime and anywhere.

II. System Overview and Objectives

The Smart Attendance System with Auto-Time Classification is a web-based attendance management system created to automate the process of recording and monitoring student attendance. The system is designed for teachers, students, and school administrators who need a faster, more accurate, and more organized way of managing attendance records. Instead of relying on manual attendance checking, the system records attendance instantly and stores all information securely in a database for future reference.

The main objective of the system is to improve the efficiency and reliability of attendance management in schools. Specifically, the system aims to automate attendance recording, reduce manual errors, classify attendance status automatically, provide organized attendance monitoring, and maintain secure data storage. Through real-time recording and automated classification, the system helps teachers and administrators save time while ensuring accurate and accessible attendance records.

III. Features Implemented

- ❖ **Authentication and User Management**
- ❖ **Attendance and Session Tracking**
- ❖ **Class and Roster Management**
- ❖ **Frontend Features**
- ❖ **Progressive Web App**

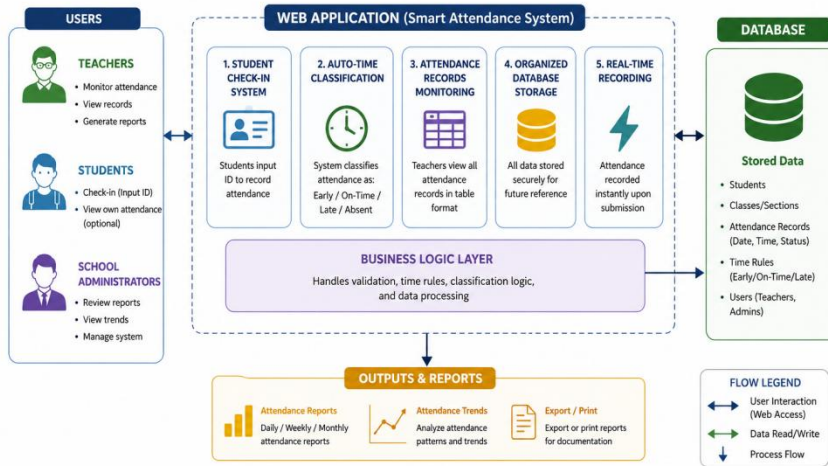


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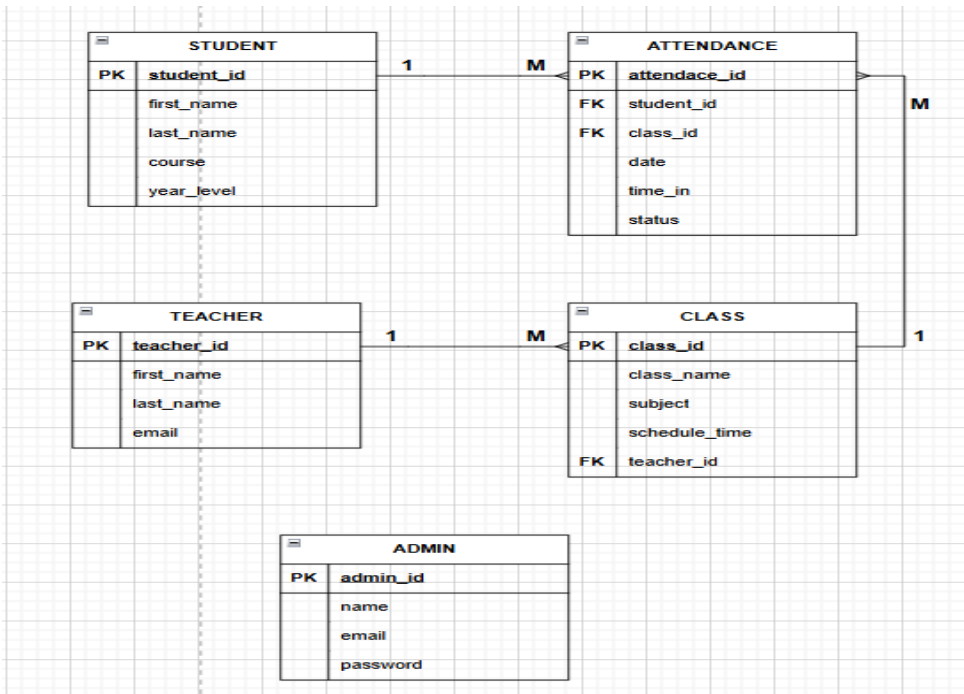


III. Architecture Diagram

Smart Attendance System with Auto-Time Classification
Architecture Diagram



IV. Entity Relationship Diagram (ERD)





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V. GitHub Repo & Commit Logs

The screenshot shows the GitHub repository interface for 'final-project--The-Survivors--BSIT2a'. The left sidebar displays the file structure with folders like 'backend', 'database', 'docs/planning', 'frontend', and 'README.md'. The main area shows the 'docs/planning' directory with a table of recent commits.

Name	Last commit message	Last commit date
..		
DFD.pdf	update	22 minutes ago
ER-Diagram.pdf	update	22 minutes ago
ProjectProposal_WST.pdf	update	22 minutes ago
USE-CASE-DIAGRAM.pdf	update	22 minutes ago

Github Repo Link

<https://github.com/alyssajenille-web/final-project--The-Survivors--BSIT2a.git>

Commit Logs:

The screenshot displays the GitHub commit log for the repository. It shows a list of commits with their messages, authors, and hashes. The commits are ordered chronologically, with the most recent at the top.

Commit Message	Author	Hash
added new documentation	Michelle Diaz committed 2 days ago	6cc58d4
fix: update live application link in README	balictarjudypear936 committed last week	d0d5b65
Merge pull request #16 from alyssajenille-web/pwa	balictarjudypear936 authored last week	e1a2381
Add PWA support with service worker, manifest, and caching strategy	balictarjudypear936 committed last week	a2b30ae
docs: update README with additional screenshots for user interfaces	balictarjudypear936 committed last week	120accd
feat: enhance README with logo, deployment link, team info, and screenshots	balictarjudypear936 committed last week	5820024
feat: enhance README with logo, deployment link, team info, and screenshots	balictarjudypear936 committed last week	5820024
Refactor code structure for improved readability and maintainability	balictarjudypear936 committed last week	4c2a772
Add Phase 4 documentation PDF for IT112 Web Systems project	balictarjudypear936 committed last week	49dc663
Remove empty .gitkeep file and add the survivors presentation PDF	balictarjudypear936 committed last week	ccdeecb
Add frontend files for Smart Attendance System	balictarjudypear936 committed last week	0fac540



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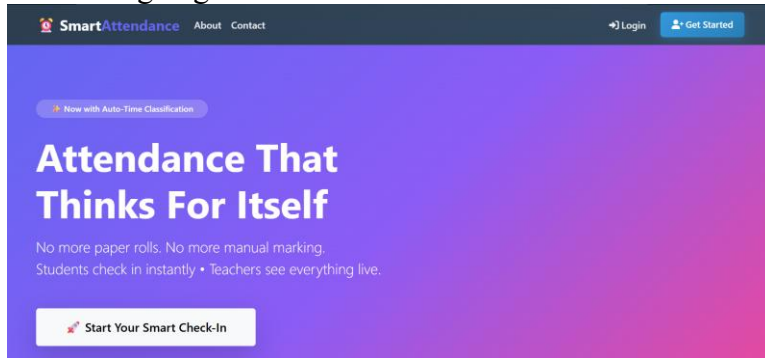


VI. Render Deployment & Live Sites

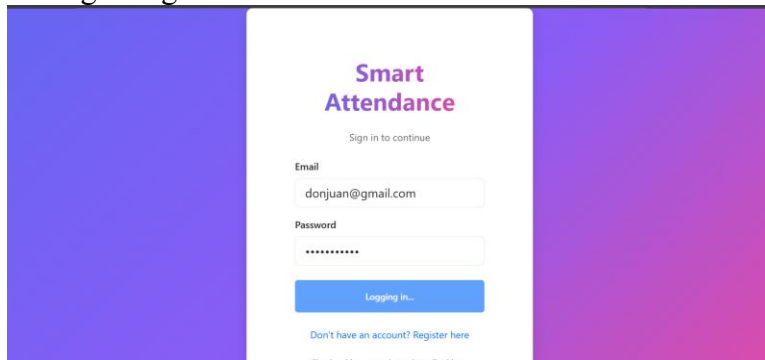
<https://final-project-the-survivors-bsit2a-1.onrender.com/>

Live Sites

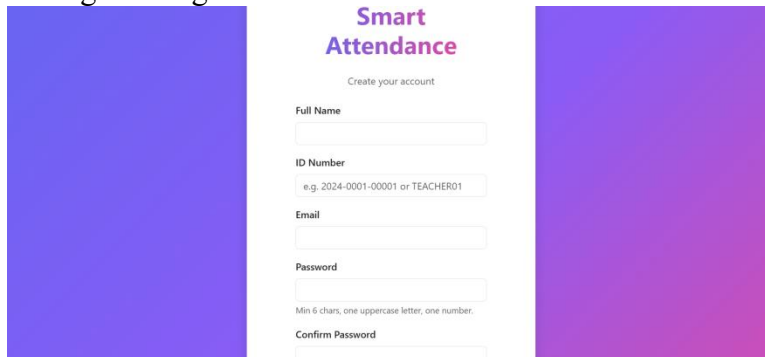
A. Landing Page



B. Login Page



C. Register Page





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D. Admin Dashboard

Welcome, admin 

ON-TIME TODAY
0

LATE TODAY
0

ABSENT TODAY
8

TOTAL TODAY
8

TOTAL USERS
13

STUDENTS
8

TEACHERS
4

Class Schedule Management

Add Class Schedule

Class Name

Class Schedules

Class	Subject	Teacher	Days	Time	Actions
IT112 - Websystems and Technologies - BSIT2A	Websystems and Technologies	Don Juan	Fri	9:00am - 10:00am	Delete

E. Teacher Dashboard

Welcome, Don Juan 

ON-TIME TODAY
0

LATE TODAY
0

ABSENT TODAY
8

TOTAL TODAY
8

[Today's Classes](#) [All Records](#) [Student Summary](#)

Today's Assigned Classes

Attendance opens and closes automatically based on the scheduled time.

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Websystems and Technologies

Today | 9:00am - 10:00am

Grace period: 5 minutes

Closed

F. Student Dashboard

Welcome, Ashely
ID: 2026-0000-00002 

ON-TIME
0

LATE
0

ABSENT
2

TOTAL RECORDS
3

Today's Classes

Your scheduled classes for today will show whether attendance is waiting, open, checked in, or closed.

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Today | 9:00am - 10:00am

Grace period: 5 minutes

Checked In

Checked In



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VII. Summary of Divisions of Tasks

Project Manager:

Throughout this project, I learned how important leadership, communication, and teamwork are in system development. Managing the group helped me improve my ability to organize tasks and monitor our progress in every phase of the project. There were many challenges during development, but we were able to solve them through cooperation and proper planning. I also learned how frontend, backend, and database components work together in a complete web application. Overall, this project improved my confidence in handling team-based projects and problem-solving situations.

Frontend Developer:

This project helped me improve my frontend development skills, especially in designing responsive dashboards and connecting the interface to backend APIs. I learned how to use JavaScript fetch(), DOM manipulation, Bootstrap components, and modals to make the system interactive and user-friendly. Creating dashboards for students, teachers, and admins also improved my understanding of role-based interfaces. There were some challenges in rendering data and handling CRUD operations but solving them improved my debugging skills. Overall, this project gave me more experience in building real-world web applications.

Backend Developer:

During this project, I learned how to create and manage API routes using Express.js and MongoDB. I improved my understanding of CRUD operations, JWT authentication, role-based access control, and attendance tracking logic. Testing API endpoints and connecting them to the frontend helped me understand how backend systems support web applications. I also learned how important clean API responses and database structure are for smooth integration. Overall, this project improved my backend programming and troubleshooting skills.

Database Manager:

This project helped me understand how databases are used in full-stack web applications. I learned how MongoDB stores attendance records, user accounts, class schedules, and session information. Monitoring database changes during insert, update, and delete operations improved my attention to detail and validation skills. I also learned how schemas affect the system's frontend and backend functionality. Overall, this project strengthened my knowledge in database management and system organization.



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GitHub Manager:

Throughout this project, I learned the importance of version control, collaboration, and proper commit management. Even though I experienced internet connection problems during some phases, I still understood how GitHub helps teams organize updates and avoid conflicts in development. Working with my groupmates taught me the value of communication and teamwork when handling project repositories. I also learned how meaningful commit messages and organized branches help maintain project structure. Overall, this project improved my understanding of collaborative software development.

Documentation Officer:

This project improved my skills in documenting workflows, organizing screenshots, and explaining technical processes clearly. I learned how to prepare reports for frontend rendering, CRUD functionalities, API integration, and database operations. Observing the development and testing process also helped me understand how each system component works together. I became more confident in creating organized reports and technical documentation for group projects. Overall, this project improved both my technical writing and debugging observation skills.